

AstroEduPlanner (AEOP): An Open-Source Educational Astronomical Observation Planner

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Software

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Summary

AstroEduPlanner (AEOP) is an open-source, Python-based tool designed to lower the technical barriers to astronomical observation planning in educational settings. Built on the Astropy (Astropy Collaboration et al., 2022; Price-Whelan et al., 2018; Robitaille et al., 2013) and Astropplan (Morris et al., 2018) libraries, AEOP enables teachers, students, and amateur astronomers to determine when and where celestial objects are observable from any location on Earth, without requiring prior knowledge of observational astronomy software or professional ephemeris systems.

The tool exposes two complementary interfaces targeting different user profiles. The first is an interactive web application built with Streamlit (Streamlit Inc., 2024), which requires no programming skills: users select their observing site, date, and targets through a graphical sidebar and obtain publication-quality altitude-versus-time plots and annual visibility calendars immediately. The second is a fully documented Jupyter Notebook that exposes the same modular Python backend, allowing students and instructors to inspect, modify, and extend the underlying calculations as part of coursework in observational astronomy or scientific computing.

AEOP resolves target names through the SIMBAD database via Astroquery, computes Solar System body positions for ten objects, estimates lunar interference (illumination fraction and angular separation), and integrates real-time cloud cover forecasts from the Open-Meteo API (Zippenfenig, 2023). The software has been developed with Brazilian public schools and higher education institutions in mind, where access to commercial observatory planning platforms is typically unavailable. Both the Streamlit interface and the Jupyter Notebook are available in English and Brazilian Portuguese, selected via a single sidebar control, reflecting the tool's dual audience of international educators and Brazilian teachers and students for whom operating a scientific tool in a foreign language would constitute an additional and unnecessary barrier. This version is also available at zenodo (Ferreira, 2026).

Statement of Need

Observation planning is a foundational skill in practical astronomy education. Before a telescope session — whether at a secondary school, a university observatory, or a public outreach event — students must determine which objects will be above the horizon, assess the interference from the Moon, check atmospheric conditions, and justify their target selection within a limited observing window. This process requires knowledge of coordinate systems, astronomical twilight, airmass, and ephemeris computation that are rarely covered in introductory courses before hands-on observing sessions begin (Bretones et al., 2016).

A gap exists for a tool that simultaneously serves two distinct educational audiences:

41 students who benefit from a visual, interactive interface requiring zero programming
 42 knowledge, and students learning scientific Python who need a structured, well-documented
 43 codebase they can explore and modify. This dual-interface design philosophy is the primary
 44 innovation of AEOP relative to existing solutions, and mirrors pedagogical strategies
 45 increasingly adopted in astronomy education research (Caballero et al., 2018).

46 The need is particularly acute in Brazil and analogous middle-income countries, where
 47 public secondary schools and regional universities face severe resource constraints.
 48 Commercial observation planning platforms are expensive or inaccessible from institutional
 49 networks with restricted outbound connectivity. AEOP is entirely free, open-source
 50 under the MIT Licence, operates without authentication, and connects only to free APIs
 51 (Nominatim for geocoding, SIMBAD for target resolution, Open-Meteo for weather). The
 52 preset target list has been curated to include southern-hemisphere showpieces — Omega
 53 Centauri, 47 Tucanae, Eta Carinae Nebula — that are prominent from Brazilian latitudes
 54 and rarely foregrounded in northern-hemisphere-centric textbooks (Bretones et al., 2016).

55 Several papers in the Journal of Open Source Education and the Journal of Open Source
 56 Software have demonstrated the value of purpose-built, open-source tools for narrowing
 57 access gaps in astronomy education (Fitzgerald et al., 2021; Villano et al., 2023). AEOP
 58 follows this tradition by targeting the specific bottleneck of pre-observation planning, a
 59 step that is consistently identified by instructors as the stage where students require the
 60 most scaffolded support.

61 State of the Field

62 Several established tools address observation planning for professional or advanced amateur
 63 use, but none combines a no-code interface with a programmable, pedagogically structured
 64 backend. Stellarium (Chéreau et al., 2023) and Cartes du Ciel provide rich sky visualisation
 65 but offer no programmatic API and cannot be scripted for classroom assignments. Skyfield
 66 provides precise ephemerides as a Python library but has no built-in educational interface
 67 and presupposes substantial programming experience. Astroplan (Morris et al., 2018)
 68 targets professional observatory scheduling and, while powerful, lacks a user-friendly
 69 interface accessible to students with no programming background.

70 AEOP occupies a distinct niche by spanning this spectrum: it pairs a no-code web
 71 interface (Streamlit) with a programmable Jupyter Notebook, both built on the same
 72 modular scientific backend. This makes it appropriate for a continuum of users from
 73 secondary-school students with no coding experience to undergraduates learning the
 74 Astropy ecosystem, an audience and use case not served as a single integrated tool by the
 75 existing software listed above.

76 Software Design

77 AEOP is organised as a modular Python package whose scientific logic is fully decoupled
 78 from its two user-facing interfaces. The backend consists of five independent modules
 79 in `src/`: `config.py` (centralised configuration and preset target lists), `location.py`
 80 (geolocation via Nominatim and coordinate handling), `targets.py` (target management
 81 and SIMBAD/Astroquery resolution with a `SkyCoord.from_name()` fallback), `analysis.py`
 82 (visibility, airmass, lunar, and Solar System computations using Astropy's AltAz frame
 83 transformations and `get_body()`), and `plotting.py` (all visualisation routines). All
 84 user-visible strings are centralised in a single `src/i18n/strings.py` module containing 62
 85 English/Portuguese string pairs, so that adding a new language is a localised, non-invasive
 86 change.

87 The two interfaces consume this shared backend without duplicating logic. The Streamlit

application (`app.py`) wraps the backend in a graphical sidebar requiring no programming, while the Jupyter Notebook (`analise_astronomica.ipynb`) exposes the same functions for inspection and extension. This separation of concerns was deliberately chosen to align with software-engineering concepts typically introduced in second-year undergraduate curricula. The project ships with a `pytest` test suite covering the analysis, location, and target modules; continuous integration via GitHub Actions; dependency management through `requirements.txt` and `pyproject.toml`; and machine-readable citation metadata in `CITATION.cff`.

Instructional Design

AEOP's architecture reflects two evidence-based instructional principles. The first is the separation of conceptual understanding from technical complexity: students who are learning *what* astronomical visibility means should not be simultaneously required to learn *how* to write Python. The Streamlit interface eliminates programming as a barrier, allowing the instructor to focus discussion on the underlying astronomy — why does Orion disappear from Brazilian skies in May? how does a 70% illuminated Moon affect observation of a faint galaxy? — rather than on software operation.

The second principle is progressive disclosure. The Jupyter Notebook is structured so that all parameters are configurable in a single cell at the top of the document before any calculations run. Students in an introductory scientific Python course can therefore begin by simply changing values (`NOME_DA_CIDADE`, `DATA_ANALISE`, `ELEVACAO_MINIMA_GRAUS`) and re-running cells, observing how outputs change before they are expected to read or modify the underlying functions in `src/`. Advanced students can then inspect `src/analysis.py` and `src/plotting.py` to understand how Astropy's AltAz frame transformation pipeline works, providing a direct pedagogical pathway into the Astropy ecosystem (Astropy Collaboration et al., 2022).

A third design principle is **linguistic accessibility**. Language barriers constitute a genuine form of exclusion in science education: Brazilian public school teachers and students with limited English proficiency have historically been unable to use English-only scientific tools without additional support. AEOP addresses this directly with a built-in bilingual interface (English and Brazilian Portuguese) controlled by a single `selectbox` in the Streamlit sidebar and by parallel `# EN: / # PT:` comments throughout the Jupyter Notebook configuration cell. The Portuguese interface allows Brazilian instructors to focus classroom time on the astronomy rather than on navigating a foreign-language tool; the English interface ensures the software is fully accessible to and citable by the international astronomy education community. The 62 string pairs are managed in a single `src/i18n/strings.py` module, designed to make the addition of further languages (e.g. Spanish, for broader Latin American reach) a localised and non-invasive change.

Features and Functionality

AEOP's core capabilities, verified against the source code in `src/`, include: (1) **nightly visibility analysis** — altitude-versus-time plots sampled at 5-minute intervals across the astronomical night (Figure 1); (2) **annual visibility calendar** — a month-by-day heatmap of observing hours for any target over a full year (Figure 2); (3) **polar sky map** — a North-up azimuthal chart of all targets above the horizon at a user-selected time, controlled by an interactive slider in both interfaces; (4) **lunar impact** — Moon illumination fraction and angular separation from any target; (5) **weather forecast** — cloud cover for the coming night via the Open-Meteo API (Zippenfenig, 2023); (6) **flexible target selection** — deep-sky objects resolved via SIMBAD/Astroquery with `SkyCoord.from_name()` fallback, plus Solar System body positions for ten objects via

136 `get_body()`; and (7) **bilingual interface (EN/PT)** — a single sidebar selector switches
137 all 62 visible strings between English and Brazilian Portuguese, with further languages
138 addable via `src/i18n/strings.py`.

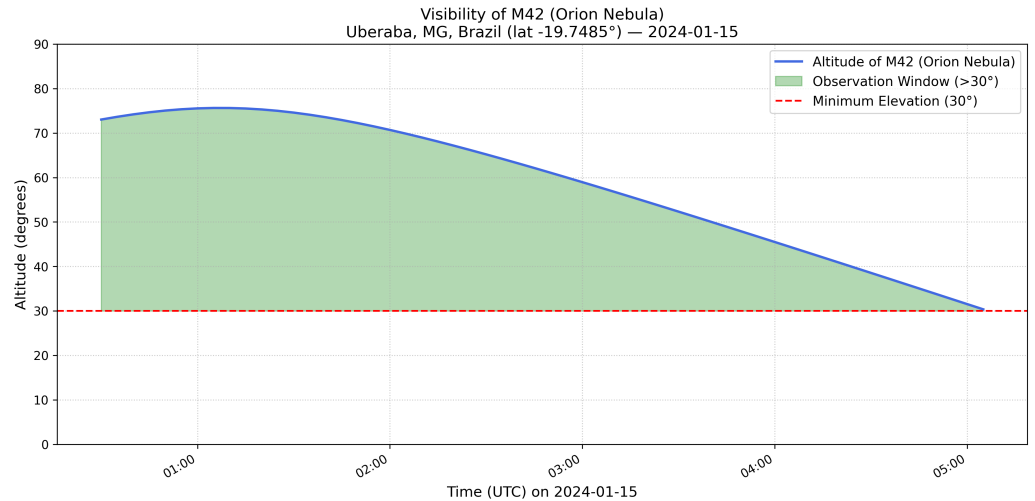


Figure 1: Altitude-versus-time plot generated by `plot_target_visibility()` for the Orion Nebula (M42) observed from Uberaba, Brazil (lat. -19.7°) on a representative winter night. The green shaded area marks the observation window above the 30° elevation threshold; the red dashed line marks the threshold itself.

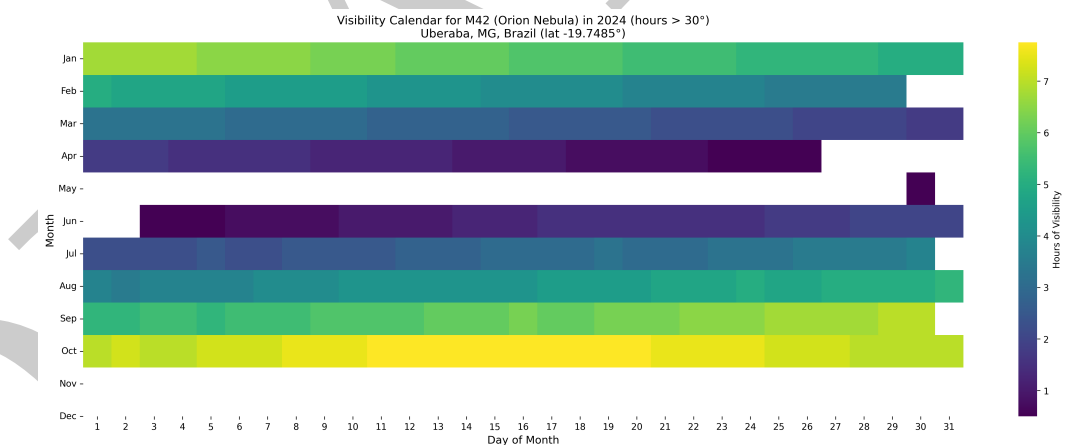


Figure 2: Annual visibility calendar heatmap generated by `plot_yearly_visibility()` for M42 (Orion Nebula) as seen from Uberaba, Brazil in 2024. Each cell encodes the number of hours the target was above 30° on that night; darker cells indicate longer visibility windows. The gap in June–July reflects the southern winter when Orion is a daytime object.

139 Usage in Teaching

140 **Scenario 1 — International university course (English interface).** In an
141 undergraduate observational astronomy course, students use the English Streamlit
142 interface to draft a pre-observation proposal before a scheduled telescope session: selecting
143 three targets with peak altitude above 45° , verifying lunar separation exceeds 30° for
144 each, and confirming cloud cover forecast is below 20%. They justify target choices by
145 annotating the altitude plots generated by AEOP. This workflow mirrors professional

proposal formats while remaining accessible to second-year students with no prior planning-software experience.

Scenario 2 — Brazilian public school (Portuguese interface, no coding required). A secondary school teacher switches the interface to Portuguese and directs students to the Streamlit application, requiring only a browser. Aligned with the Brazilian National Curriculum (BNCC) competencies on Earth’s position in the Solar System, the activity asks: “In which month can we best observe the Southern Cross from here? What would change if we were in Manaus instead of Porto Alegre?” Students adjust parameters and compare altitude plots with no code involved. The fully Portuguese interface eliminates the foreign-language barrier that would otherwise prevent independent use in this context.

Scenario 3 — Scientific Python course (bilingual Jupyter Notebook). Students open `analise_astronomica.ipynb`, where the configuration cell contains `parallel # EN: / # PT:` comments on every line. They begin by modifying only the configuration variables, then read the English-documented source in `src/analysis.py` to understand how Astropy’s AltAz frame transformation works, and finally write a short function that extends the existing pipeline. The bilingual structure allows native Portuguese speakers to start in their first language and transition to English documentation as their confidence grows.

Research Impact Statement

AEOP’s intended impact is on astronomy and scientific-computing education rather than on the production of new research data. By removing the technical and linguistic barriers to pre-observation planning, the tool enables instructors to devote scarce classroom and observatory time to astronomical reasoning instead of software operation. The dual-interface design supports a deliberate learning progression — from no-code exploration to inspection and extension of a documented Python codebase — that scaffolds students into the broader Astropy ecosystem and into general scientific-software practices such as modular design, unit testing, and version control.

The software is expected to be most impactful in under-resourced contexts, particularly Brazilian public secondary schools and regional universities, where commercial planning platforms are financially or technically inaccessible and where an English-only interface would itself exclude many learners. By foregrounding southern-hemisphere targets and providing a fully Portuguese interface, AEOP aims to make practical observation planning attainable for educational communities that existing tools have not served, while remaining fully usable and citable by the international astronomy-education community in English.

AI Usage Disclosure

Generative AI tools were used in a limited, assistive capacity during the preparation of this work. AI assistance was used to help draft and copy-edit portions of the manuscript text and software documentation, and to suggest refactorings and docstrings during code development. All scientific content, the software architecture and algorithms, the test suite, and all final text were designed, written, reviewed, and verified by the authors, who take full responsibility for the correctness and integrity of the software and the paper. No AI-generated content was incorporated without author review.

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